America Technical Center

AMERICA TECHNICAL REPORT

ATR NUMBER:	26-008	**PROJECT NO.:**		**DATE:**	7/28/2026
ORIGINATOR:	Shaun Sekura	**MANAGER:**	Bob Wheeler		
PROJECT:	Summer 2026 V&VT Internship				
REPORT SUBJECT:	End of Internship Report				
KEYWORDS:	Cord, Compound, Bimodal, MTS-810, Abaqus, T360				
cc:	ATC Web, ARENA				

Executive Summary

Summary:

This session included 4 projects. The first project objective was to study the current methodology used to assign moduli to cords and compounds in Hankook's database, then validate the use of a more accurate modulus calculation to produce more accurate performance predictions with FE models. The modulus data from nylon and polyester cords was first specifically used for finite element (FE) virtual inflated shape accuracy study. The current method of calculating cord modulus is a linear regression slope of a full stress-strain curve, an engineering assumption that the stress-strain curve is linear to simplify calculations. The resulting new data produced a nonlinear cord modulus that was lower than the current linear version for both the nylon and polyester. To analyze the physical data, a second project was used in which a custom T360 surface measurement data processor was created in Python. The physical and virtual inflated profile data at 180 kPa were compared. Then the new cord data was used in the FE model and a new virtual profile was predicted. An average improvement of 50.4% at tire crown ribs and groove bases. For the second part of the first project compound bimodal moduli were reviewed to understand the logic behind the derivation of modulus from complex harmonic stress-strain testing and find any additional nonlinearities to consider for future implementation. Along with these two projects, two smaller projects included a physical tire footprint camber validation investigation and a wheel CAD FE meshing and mode shape study.

Forward Plan:

Additional research may be required to determine conclusions of improved modulus at the side of the tire profile. The T360 processor program made by me only works for the top half of the tire profile and causes an intersection with the VTA profile at the side of the tire profile. A better T360 processor could be used to further test this conclusion. If this conclusion is proven right, a refined cord database would allow for more accurate profiles.

Report:

Presentation file is attached

Shaun P. Sekura

Hankook Tire & Technology, R&D Staff Office
End of Internship Presentation
Shaun Sekura

America Technical Center
July 28, 2026

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1) Introduction

Education

Rising Junior at The Ohio State University

- Bachelor of Science in Mechanical Engineering (12/27)

THE OHIO STATE UNIVERSITY

COLLEGE OF ENGINEERING

Work Experience

Summer 2026 Co-op with Hankook Tire & Technology

- Worked in Virtual & Validation Department under Bob Wheeler
- Returning to Ohio State to continue studies
- Previous experience with EcoCAR and Formula SAE teams at Ohio State

Hankook driving emotion

EcoCAR EV CHALLENGE OSU WU

formulabuckeyes

Summer 2026 | Fall 2025 – Spring 2026 | Fall 2024 – Spring 2025

2) Development / Misc. Projects

ATRs Read:

ATR 08-024: Material Characterization for Rolling Tire Modal Analysis

ATR 04008: Material Characterization at Modal Test Strain Levels

ATR 09-041: PCI Modeling Accuracy Improvement

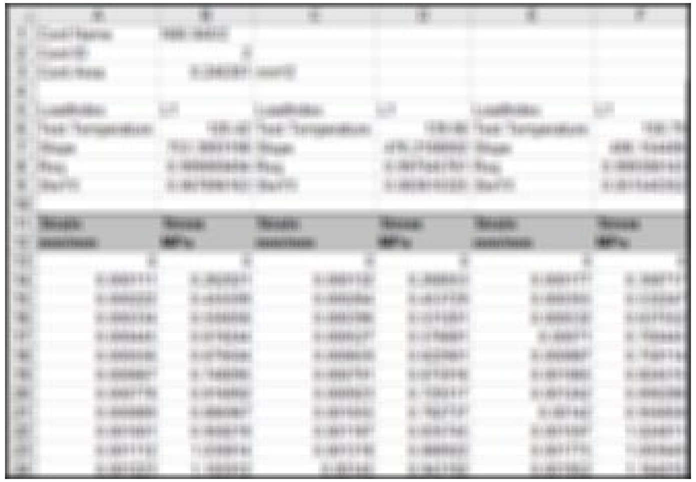
ATR 16-015: Previous Intern Project

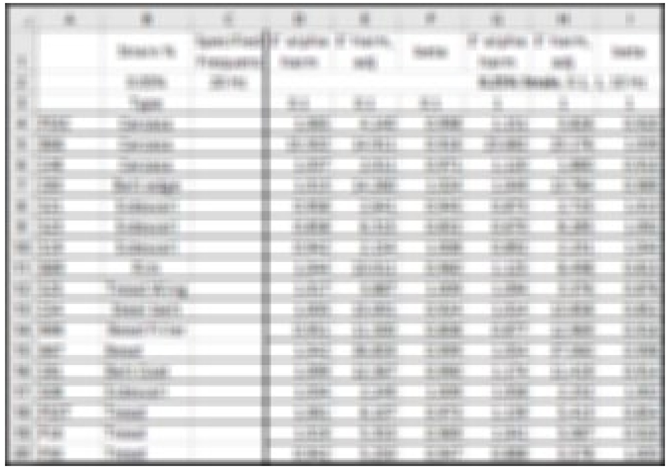
- Tutored on Abaqus, Hypermesh, and Python for virtual applications
- Tutored on Siemens Test Lab (free-free & fixed free modal testing) and FPM Machine for physical applications
- Read ATRs to understand previous work and project writing

Cord Datasheet

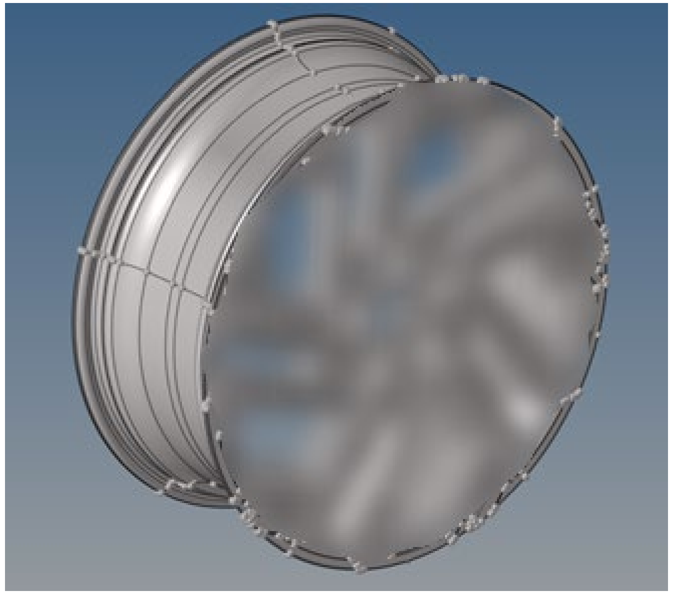
Compound Datasheet

Most ATRs were read to supplement the cord and bimodal compound project and understand where the data came from.

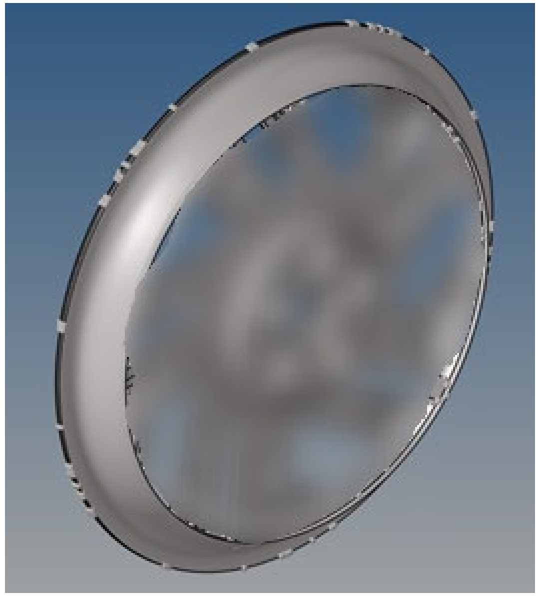




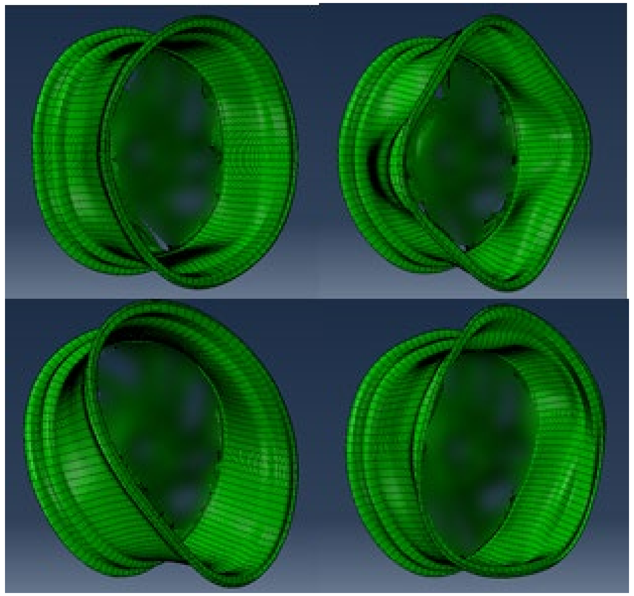
2) Development / Misc. Projects



❖ Full CAD Wheel



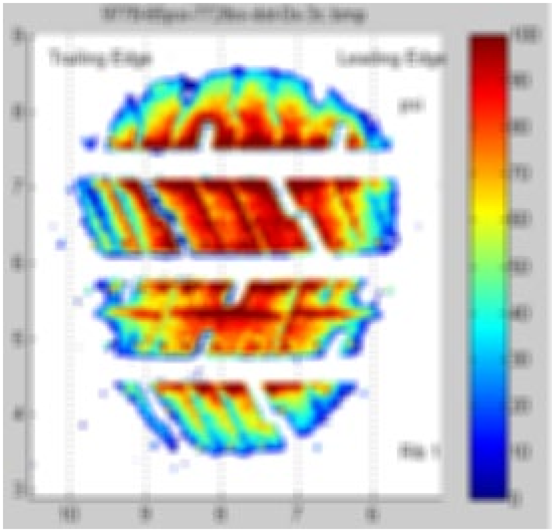
❖ Trimmed CAD Wheel Web

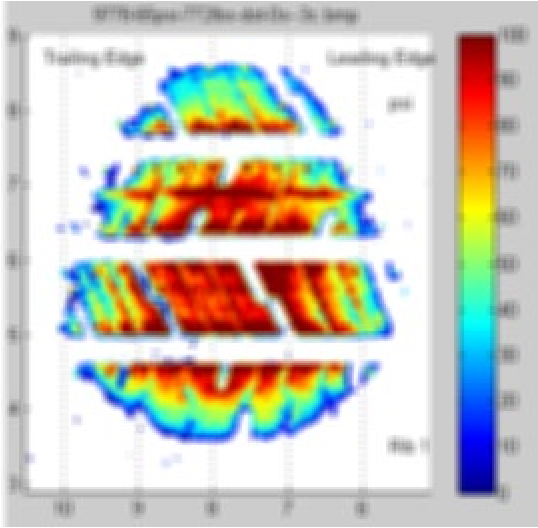


❖ Frequency Mode Shapes
w/ wheel fixed by bolt holes

- Virtual applications using OE Wheel
- Used Hypermesh to trim wheel and mesh wheel within tolerances
- Used Abaqus to apply loads and frequency to observe mode shapes

2) Development / Misc. Projects





- Footprint pictures taken with FPM Machine
- Analysis done with FP Characterization program

- Data taken at combinations of
  - 350 kg, 678.1 kg, 863.2 kg, 1360 kg
  - 60 psi, 65 psi, 70 psi, 80 psi
  - 3 degree camber, -3 degree camber

- Max lengths percent difference: 1.951%
- Average lengths percent difference: 3.033%

- Confidence in symmetrical camber being interchangeable

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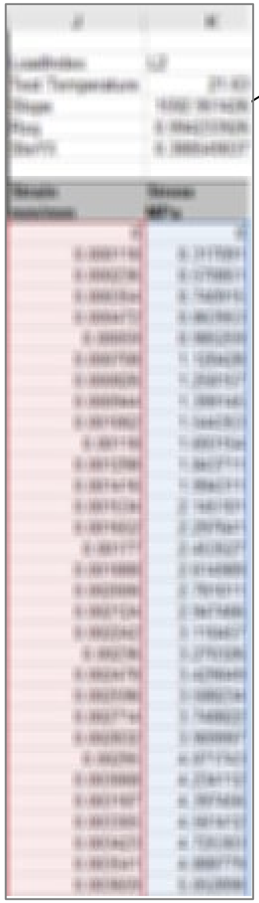
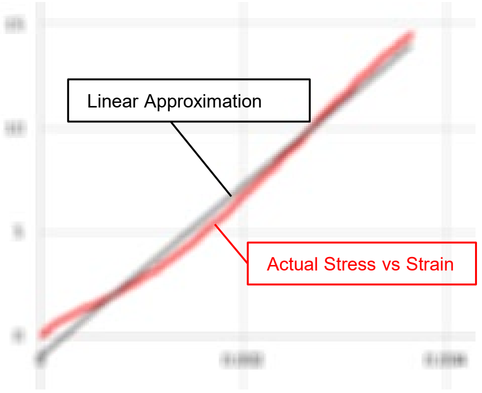
3) Cord and Bimodal Compound Project

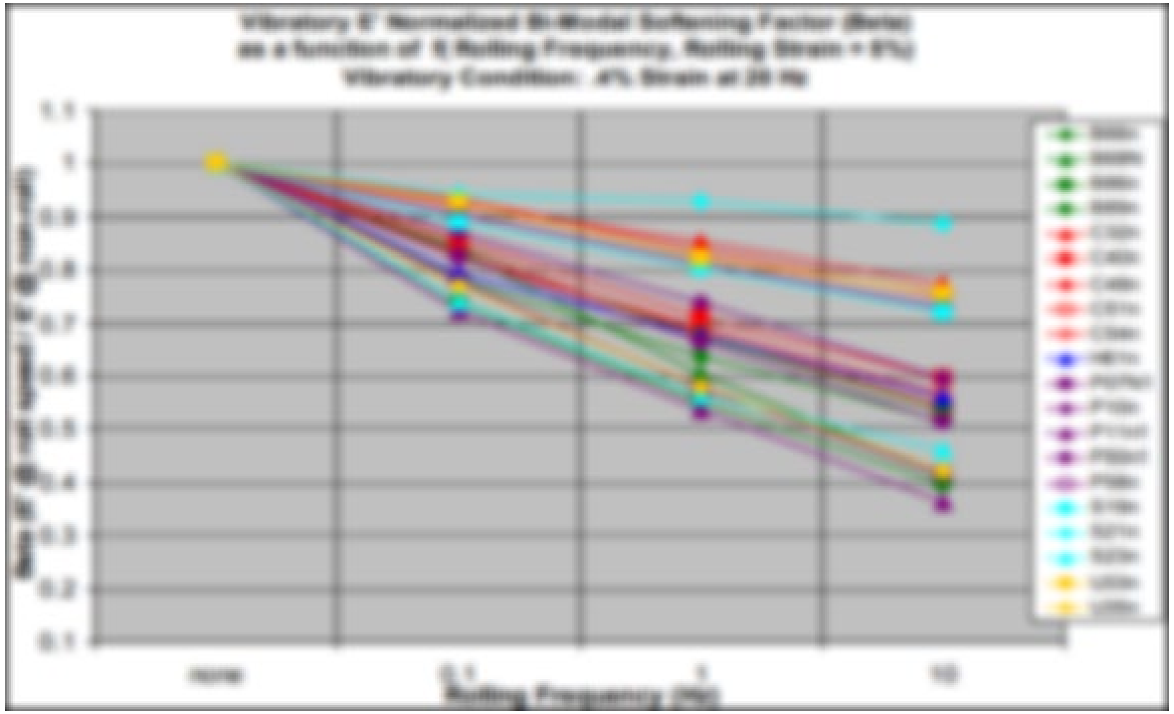
Cord



- Current method finds modulus by taking the linear regression slope
- Good method for linear analysis, modulus is approximately linear, but still nonlinear
- Set up method to adjust modulus based on given strain

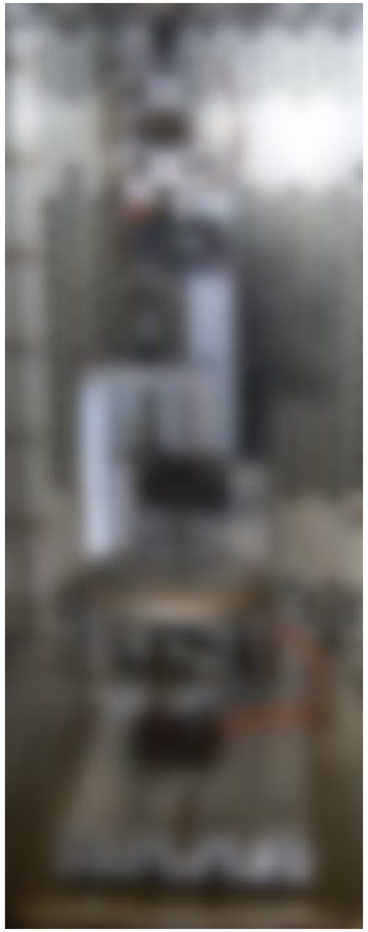


Bimodal Compound



- Current method samples modulus only at one rolling freq. at one vibratory strain at one harmonic freq. & one base strain
- Set up method to adjust modulus based on rolling freq. and vibratory strain

3) Cord and Bimodal Compound Project



Cord Creep Test Stand

Cords with tests at room temperature (~20˚ - 30˚ C) were selected for this study

Nylon

Dimensional Stable Polyester

Each cord was analyzed at 2 load ranges

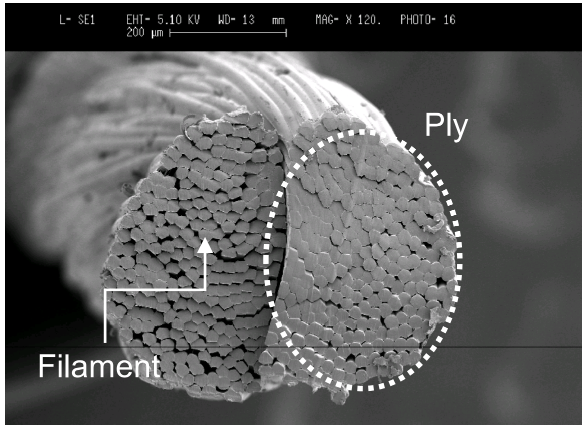
Material

Linear density

Low Twist (# of turns per unit length)

Unit of linear density (Denier)

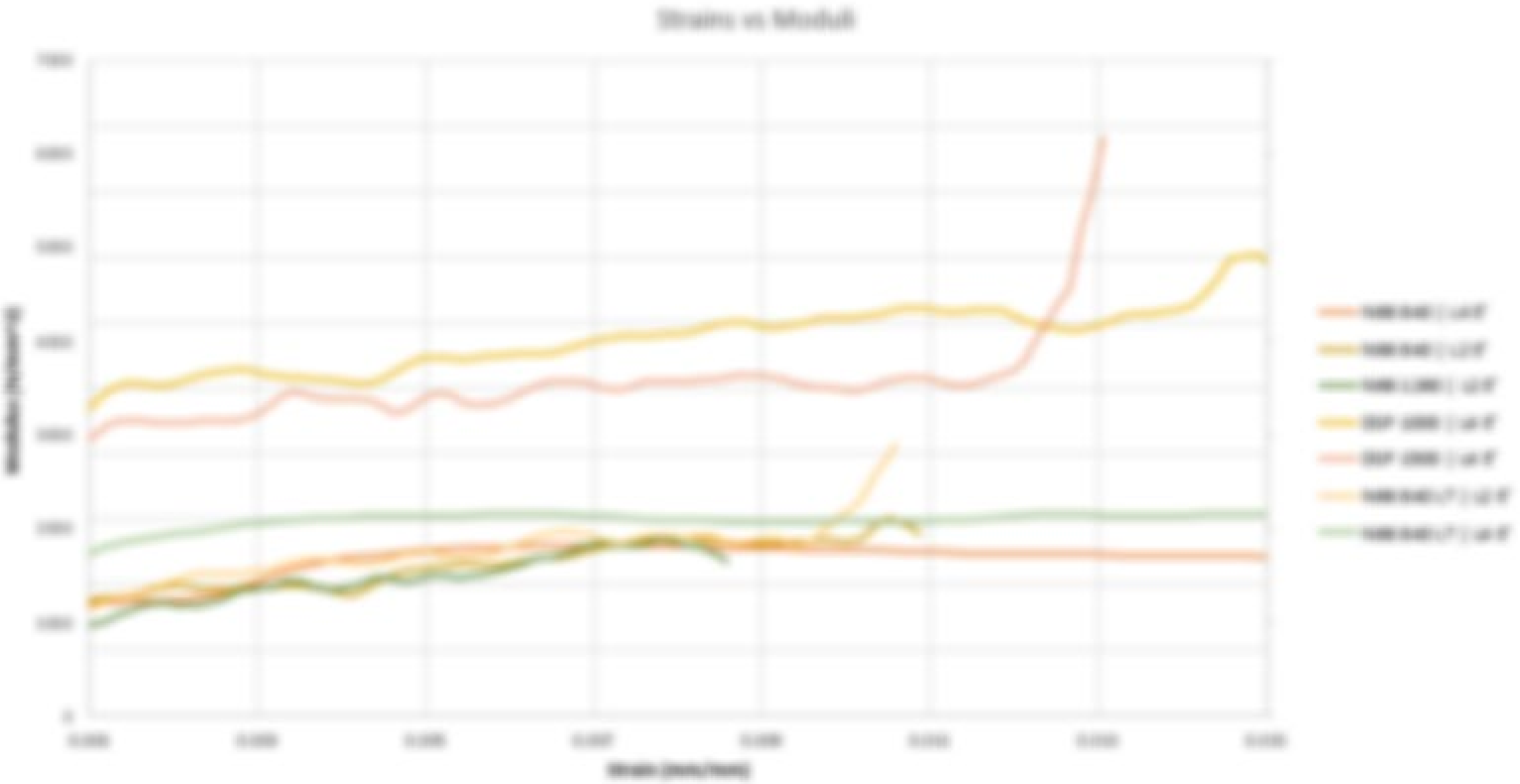
Number of ply (yarn)



300~400ea/ply

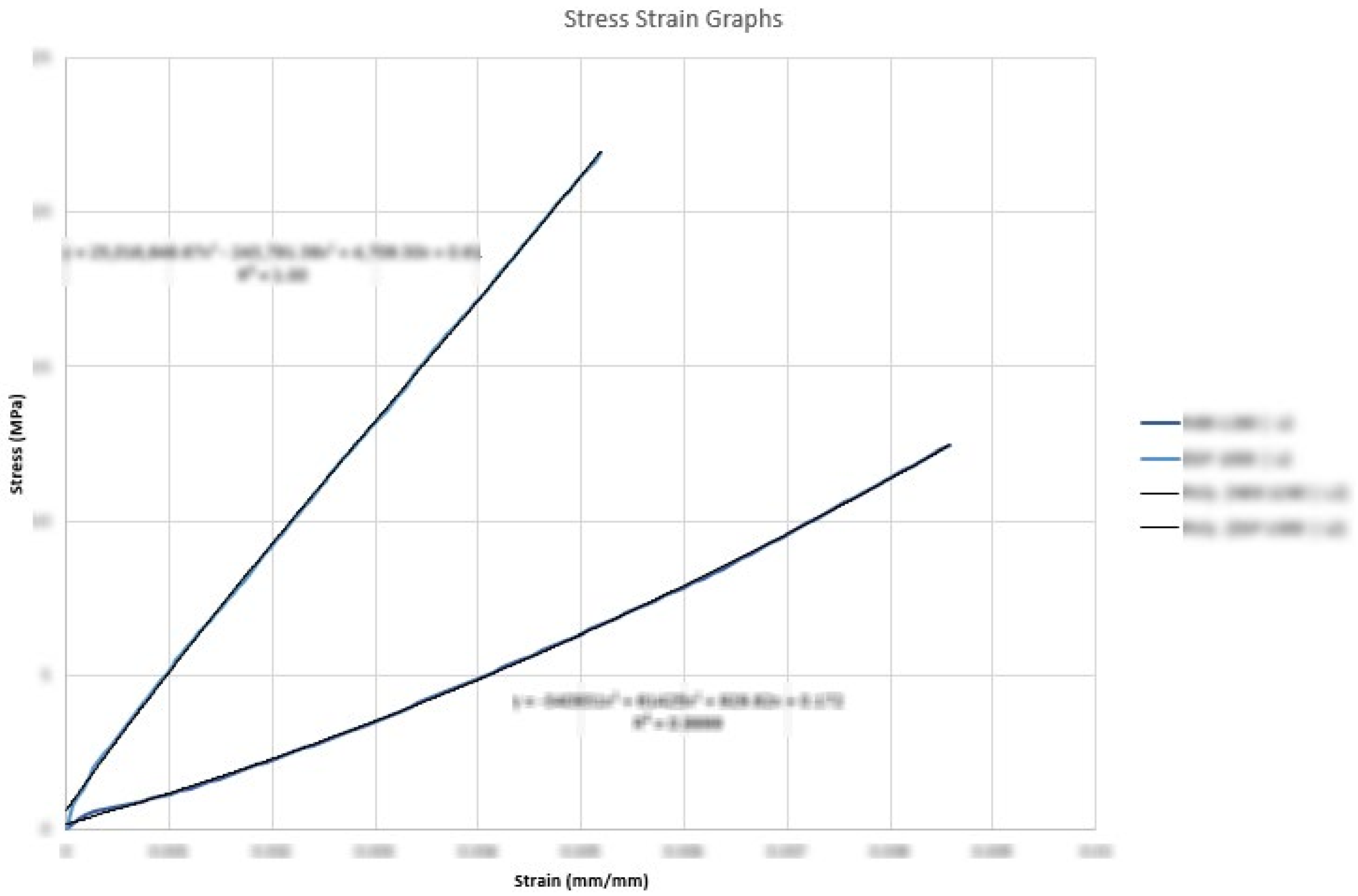
3) Cord and Bimodal Compound Project

1. Least squares linear regression slope of stress strain to calculate strain vs. moduli, noisy data

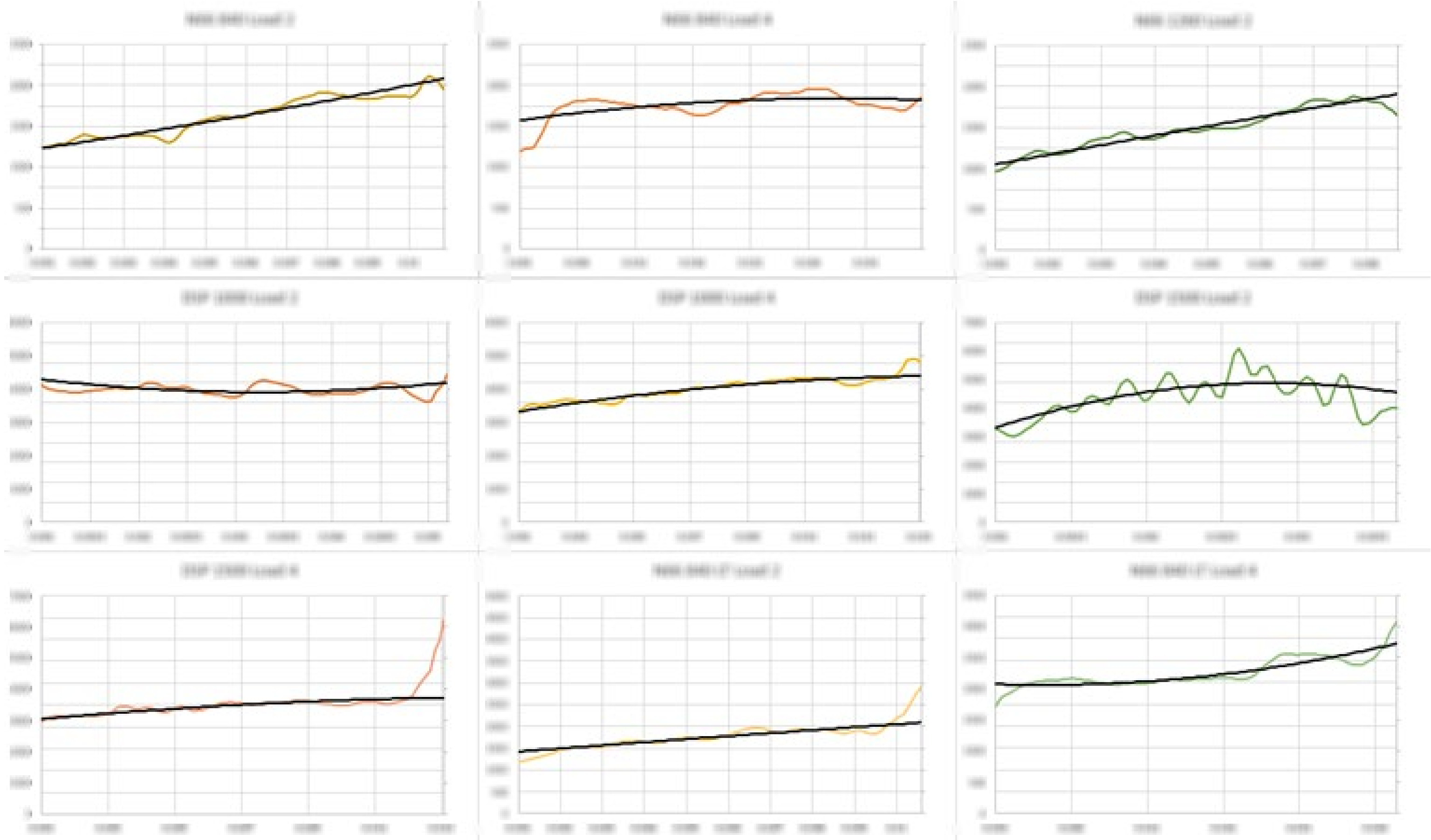


3) Cord and Bimodal Compound Project

2. Trend line to curve fit stress strain graphs



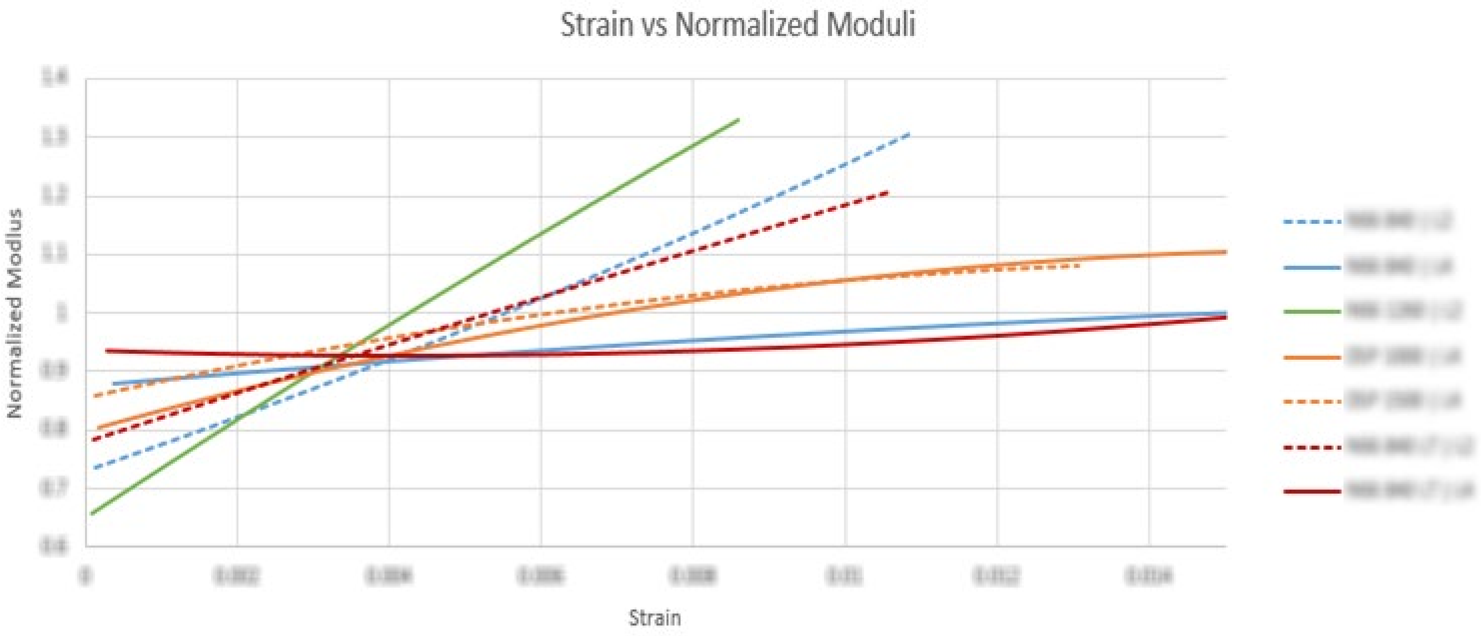
3) Cord and Bimodal Compound Project



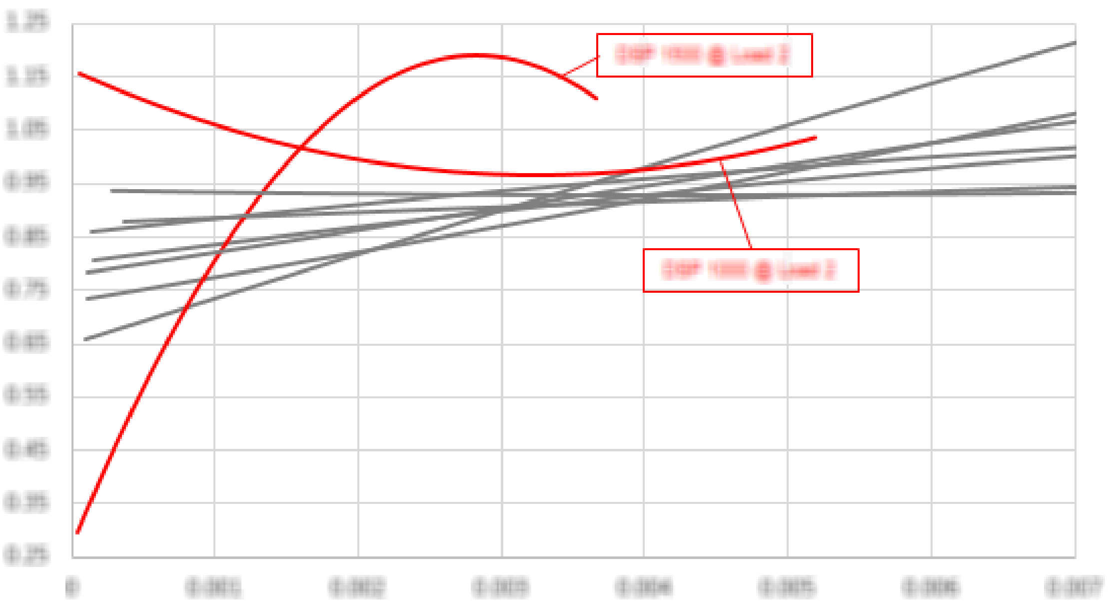
3. Take derivative of trend line to curve fit against moduli

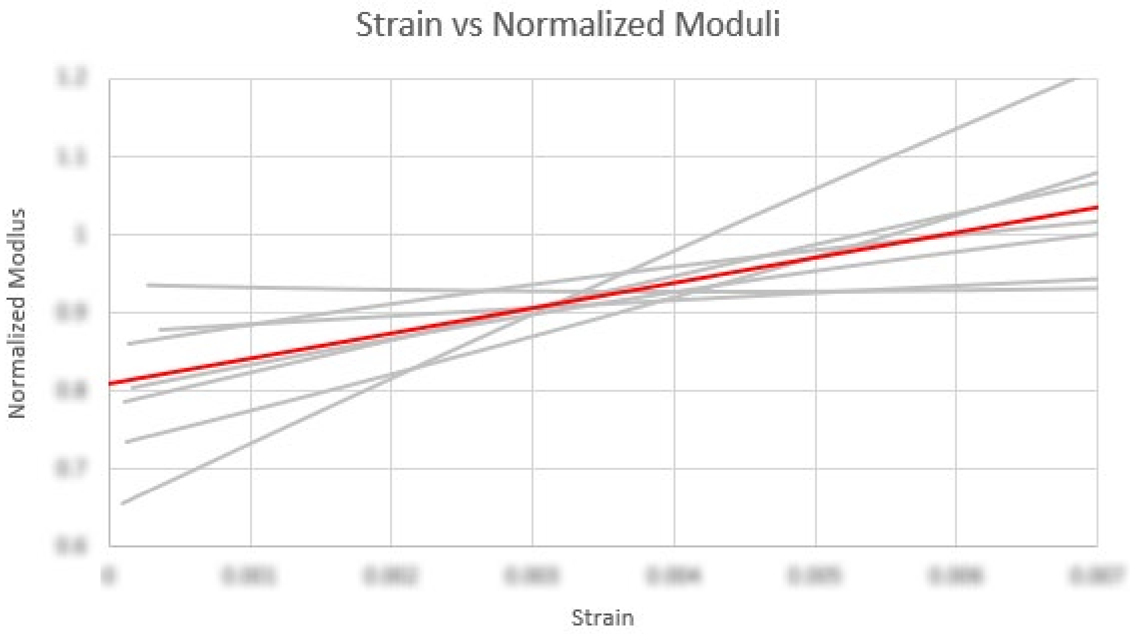
3) Cord and Bimodal Compound Project

4. Normalize about original constant modulus for comparison



3) Cord and Bimodal Compound Project





Dimensional Stable Polyester at lower loads (in red) were excluded

Behavior caused by nature of dimensional stability

- Lower loads: heat-set structure & internal stresses affect strain vs. moduli relationship
- Higher loads: stabilizing treatment is overcome and behavior follows trend

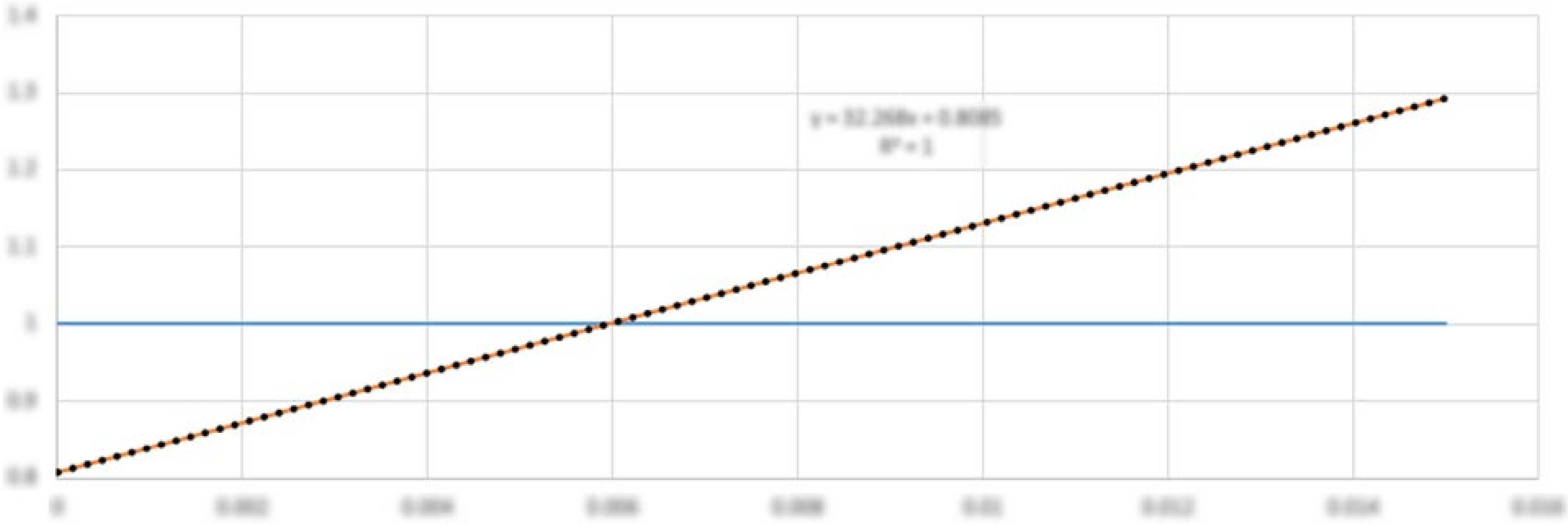
Average of strains vs moduli

Establishing a relationship between percent strain and modulus

- FORECAST.LINEAR() to interpolate values
- AVERAGE() to take average at each interpolated point

Nylon and polyester modulus samples were combined due to similarity of normalized modulus

3) Cord and Bimodal Compound Project



For averaged slope of strain vs moduli graphs:

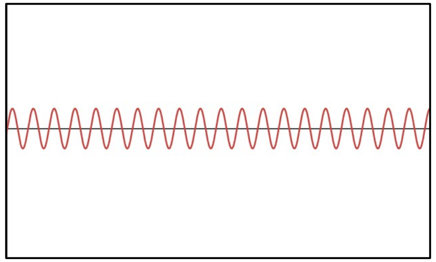
- At 0.1% strain, modulus is on average 15% lower
- At 1.0% strain, modulus is on average 15% higher

New Proposed Method: Use equations for each cord for more accurate modulus

Next Steps: apply to cord database to expand the data, apply to more cords to observe relationship

3) Cord and Bimodal Compound Project

Single sine



Single sine		1				
Condition	Amplitude	Hz.	K*	K'	K"	TanD
0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
5		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
6		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
7		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
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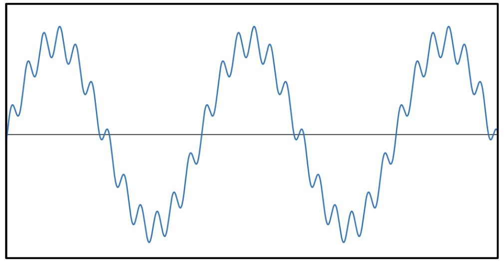
9		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Sine-on-sine



Sin-on-sine		low.[illegible]				
Condition	Amplitude	Hz.	K*	Harm K*	Phase	H Phase
0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
5		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
6		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
7		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
8		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Dynamic Stiffness Amplitude: K* = $\frac{F_o}{\delta_o}$; total stiffness of compound at test frequency

Storage Stiffness: K' = K* $\cdot \cos(\varphi)$; elastic stiffness, stores and releases energy

Loss Stiffness: K'' = K* $\cdot \sin(\varphi)$; inelastic stiffness, dampens/dissipates energy

Dynamic Stiffness Phase (φ); ratio of loss to storage stiffness

Dynamic Stiffness: K* = $\frac{F_o(f_1)}{\delta_o(f_1)}$; stiffness measured at primary (slow) frequency

Harmonic Dynamic Stiffness: Harm K* = $\frac{F_o(f_2)}{\delta_o(f_2)}$; stiffness measured at fast frequency

Phase (φ_1); time lag/response delay to loading at slow frequency

Harmonic Phase (φ_2); time lag/response delay to loading at fast frequency

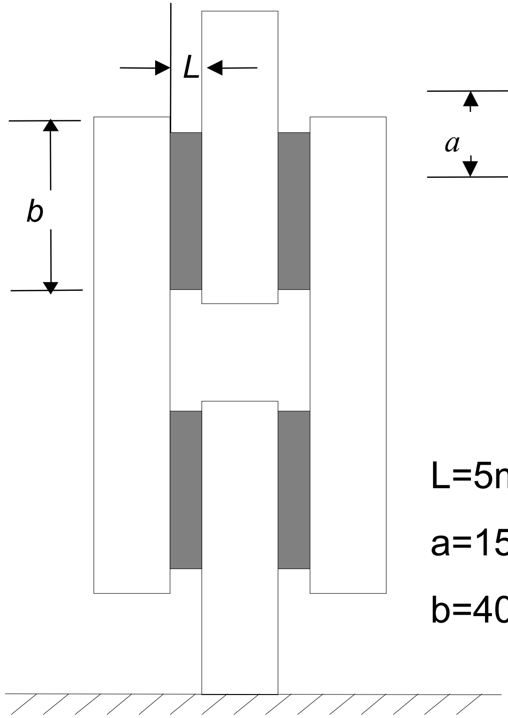
- Storage Stiffness (K') from single sine data
- Dynamic Stiffness Amplitude (K*) from SoS data

⇨

Mono-modal vibratory modulus $(\overline{E'_{mv_i}})$ and Alpha (α)

Bi-modal vibratory modulus (E'_{bv,adj_i}) and Beta (β)

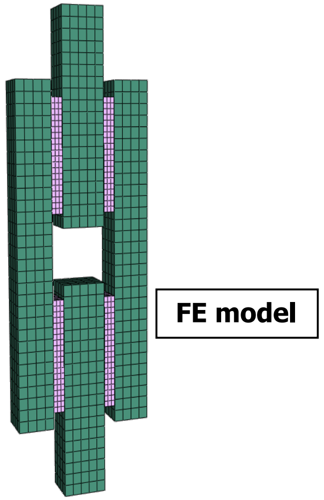
3) Cord and Bimodal Compound Project



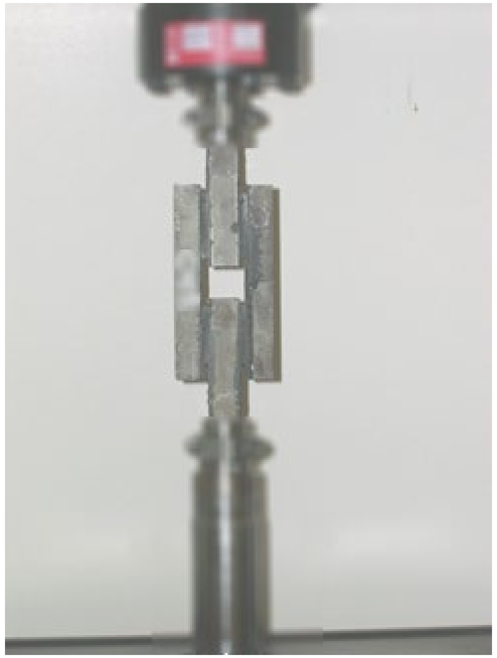
L=5mm

a=15mm

b=40mm



❖ Image from previous 2016 study



thickness

Dynamic Stiffness Amplitude
(from Sine on sine data)

Shear modulus to young modulus conversion

$$G = \frac{E}{2(1+v)} \rightarrow E = 3G$$

$$E'_{bv} = K^* \cdot \cos(\theta \cdot \frac{\pi}{180}) \cdot \frac{5}{40 \cdot 15} \cdot 0.175127 \cdot 3 \cdot \frac{1}{4}$$

Phase (from Sine on sine data)

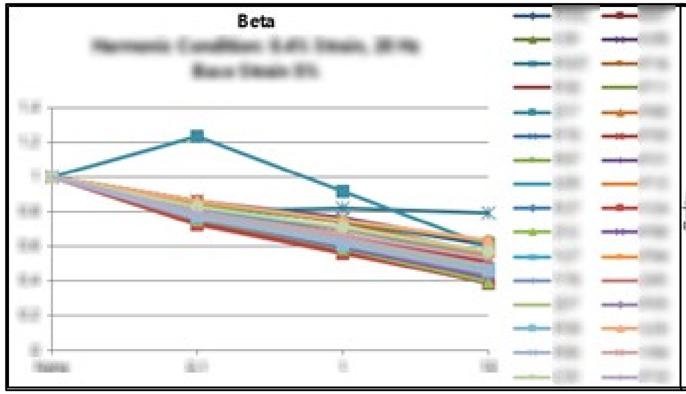
length

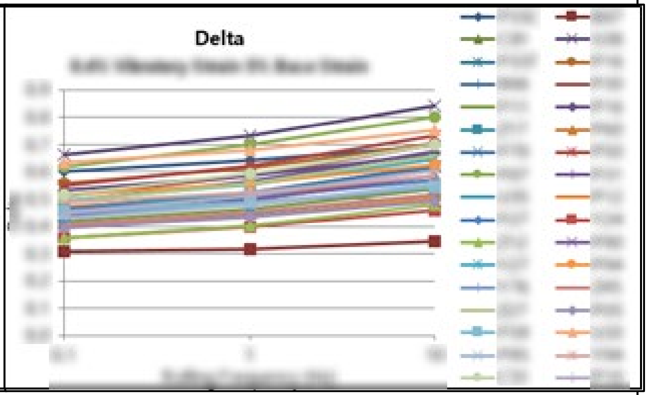
width

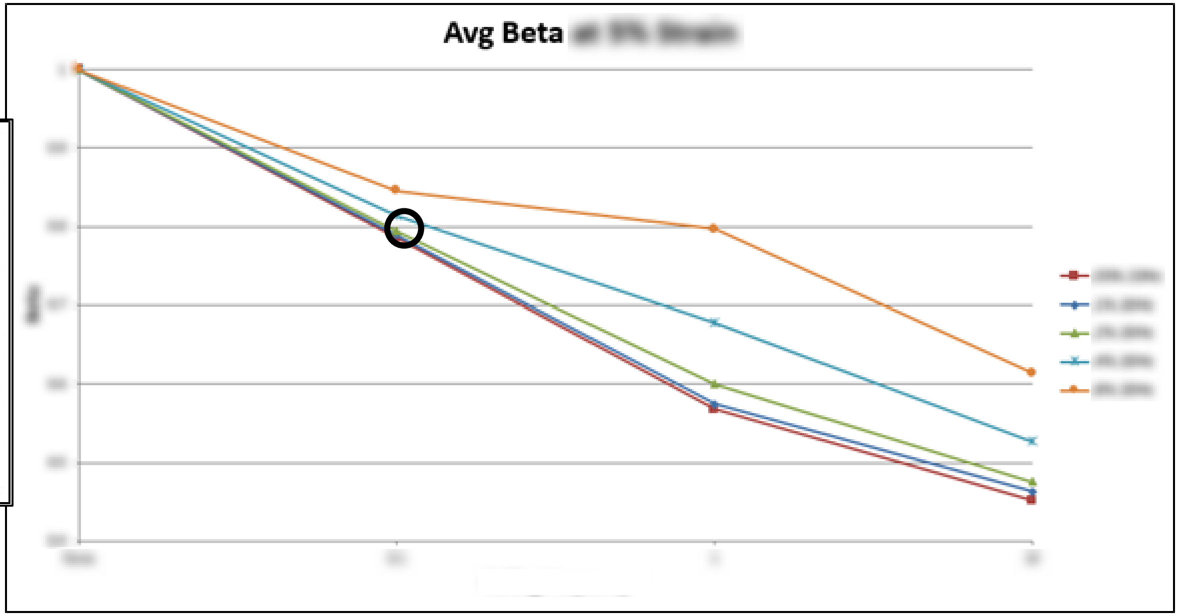
$$\frac{N}{mm} = \frac{lbs}{in}$$

quad lap

3) Cord and Bimodal Compound Project

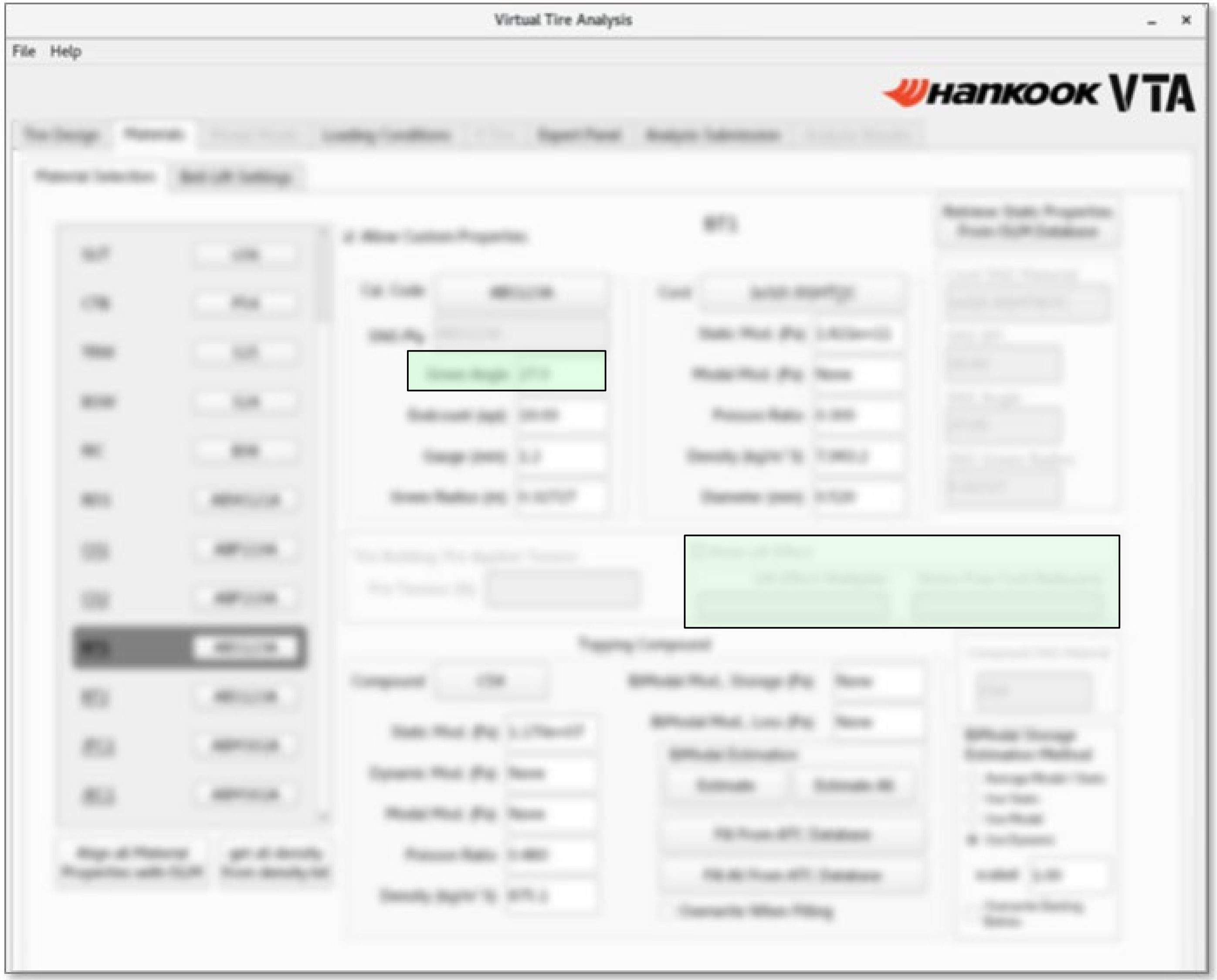






- Datasets exist from two prior studies (2007, 2016 – above data)
- Current FE process only samples at one rolling frequency, one vibratory strain & frequency
- 2016 data also has lower harmonic strain, indicates more significant beta
- New method: select rolling frequency & vibratory strain, provides more accurate modulus

4) Inflated Tire P-V Analysis



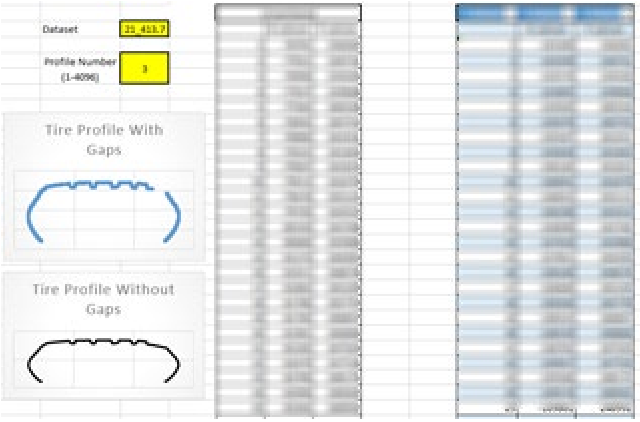
Physical vs virtual comparison of GM 9BUX Tire at 180 kPa, 27˚ Belt Angle

- Physical: T360 Tire Scanner
- Virtual: VTA (no mold lift)

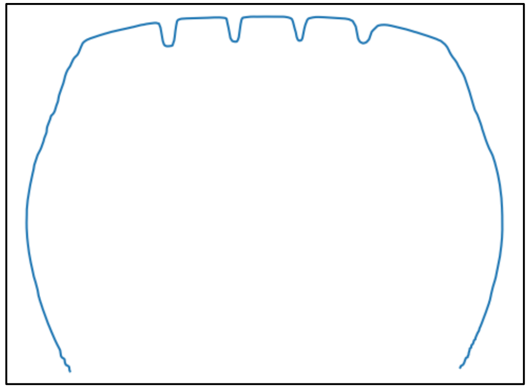
❖ OE Dev Tire on a OE Wheel during Free-Free test – Allen Basker

4) Inflated Tire P-V Analysis

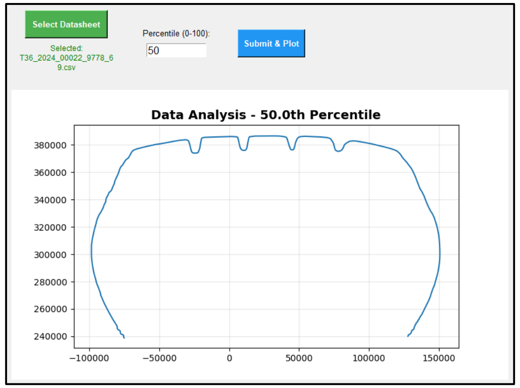
Excel to visualize data



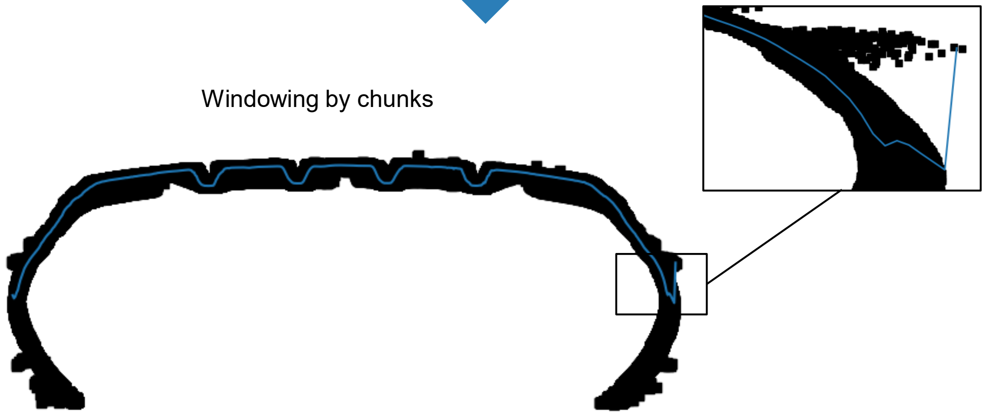
50th percentile of rows



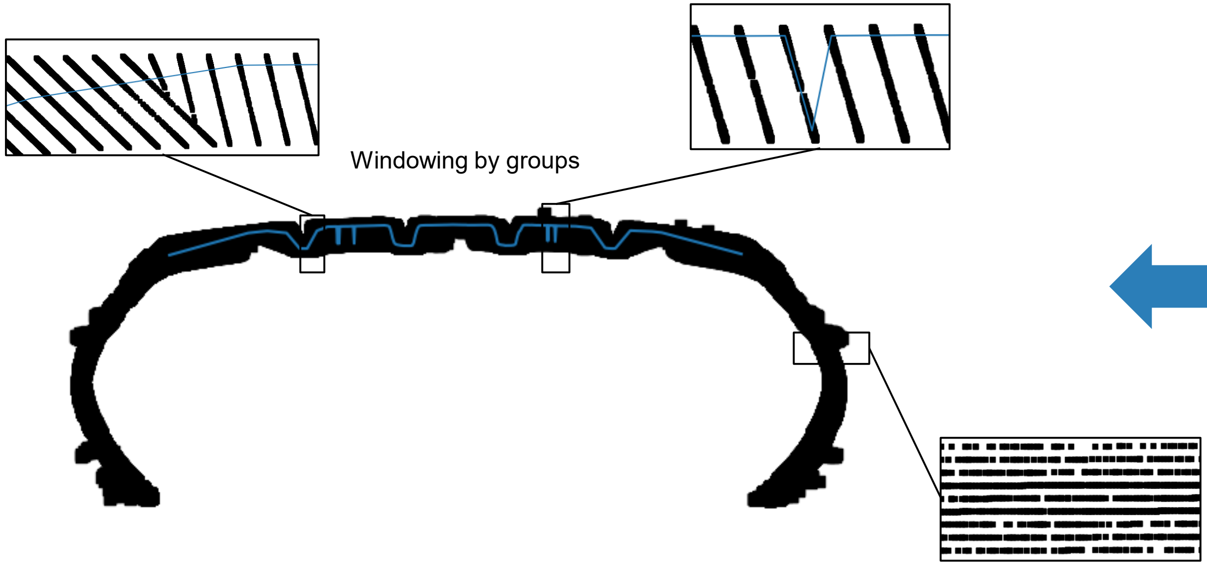
Created window



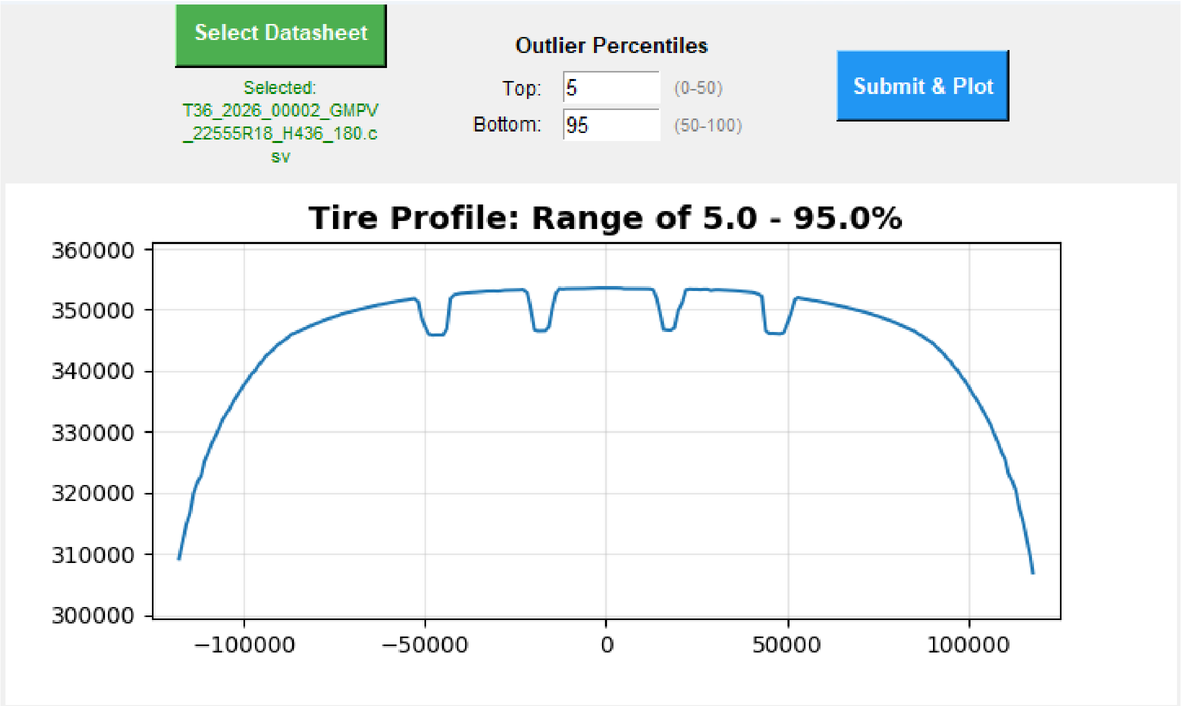
Windowing by chunks

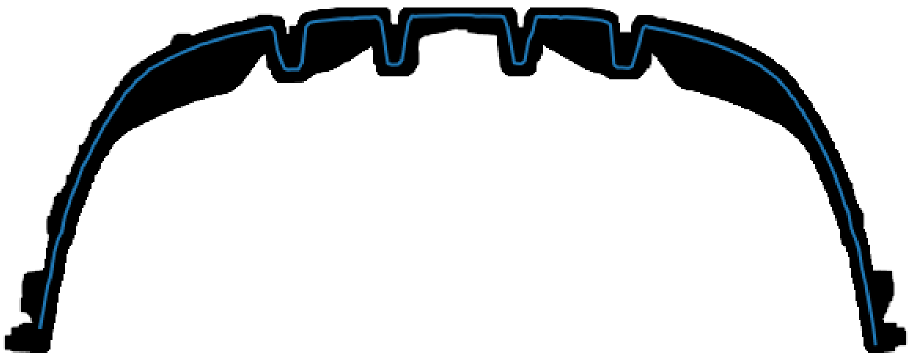


Windowing by groups



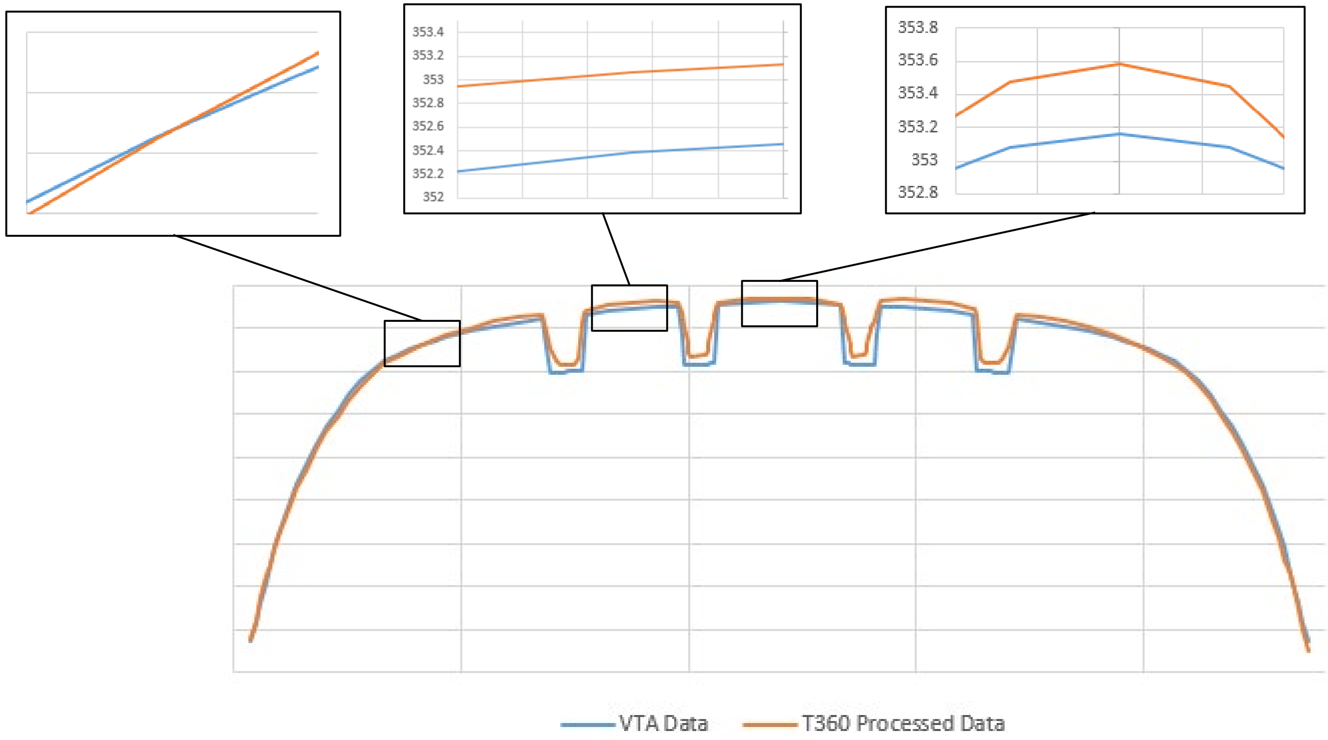
4) Inflated Tire P-V Analysis

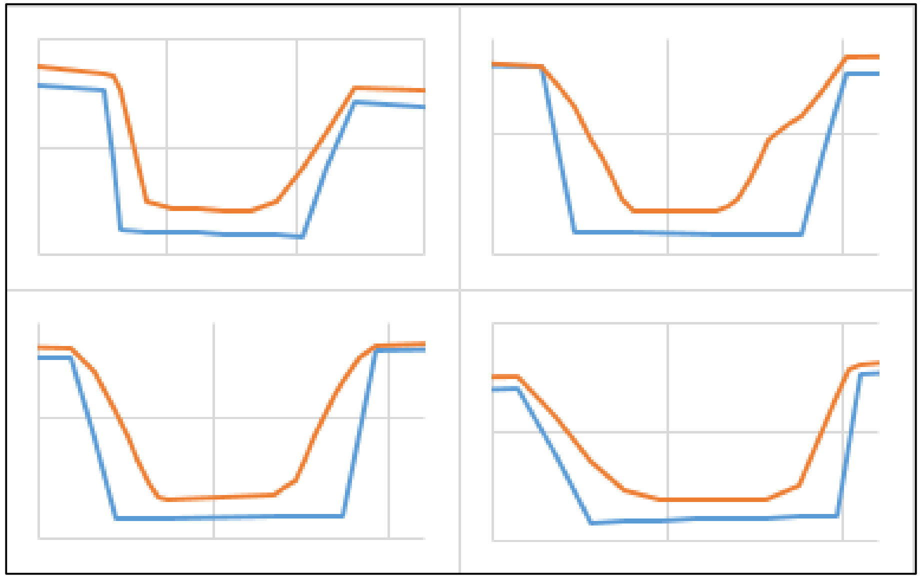




- Windowing by chunks and groups
- Centered at zero to match other files
- Modifiable outliers for fine tuning
- Searchable y values from x with interpolation for analysis against virtual files

4) Inflated Tire P-V Analysis





Total Run Time: 71.49 sec

Windowing by groups: 66.97 sec

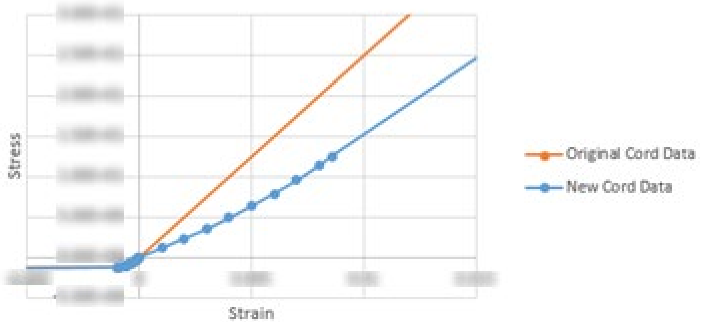
Average Deviation at Crown: 0.862 mm

Average Deviation at Shoulder/Sidewall: 0.514 mm

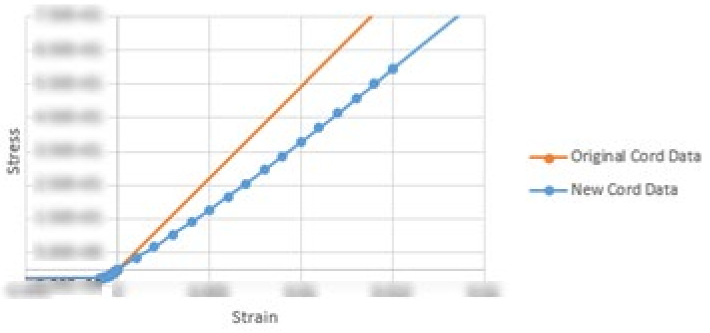
Average Deviation in Grooves: 2.853 mm

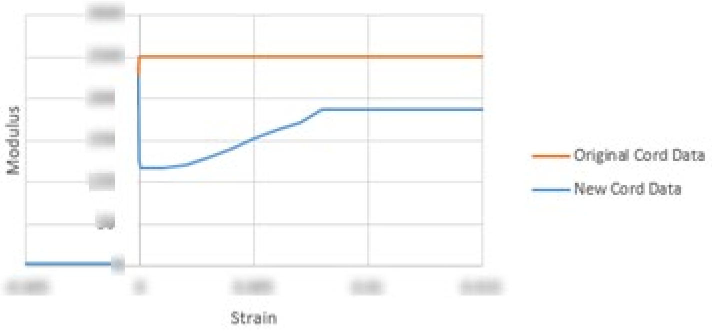
4) Inflated Tire P-V Analysis

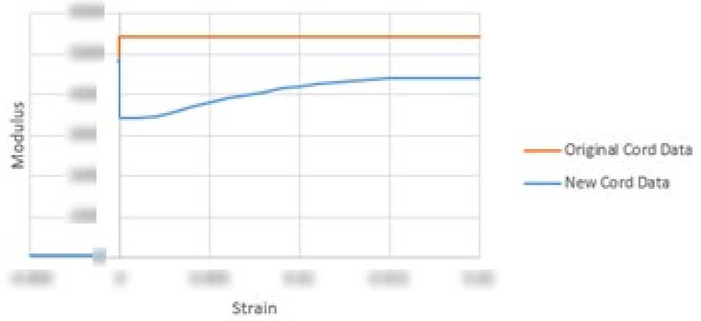
Nylon



Polyester



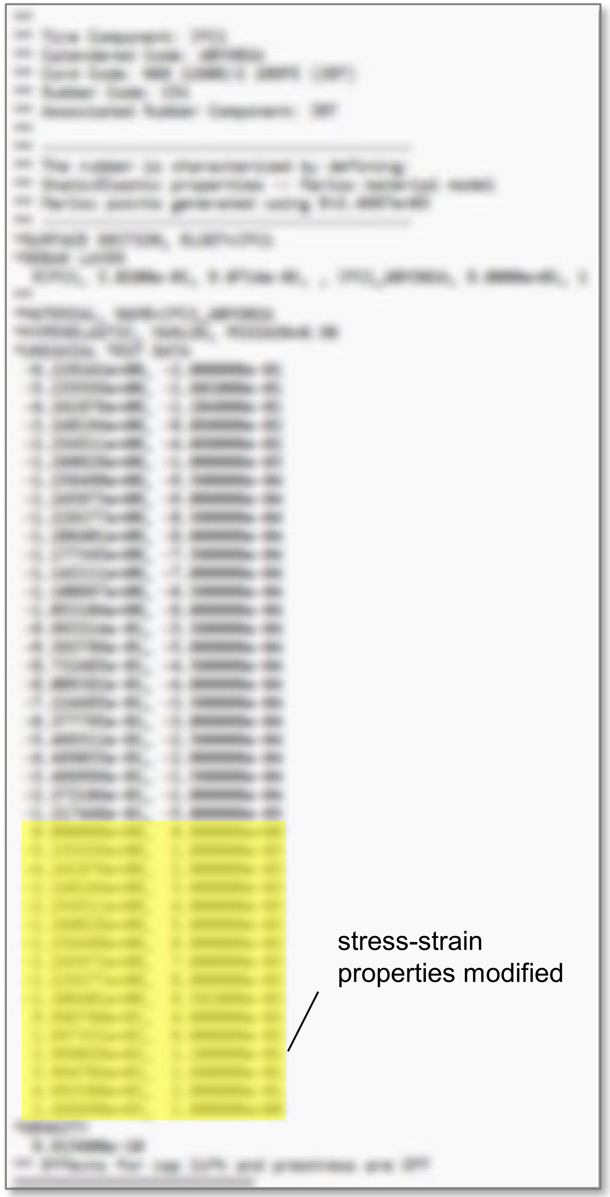




Modified input files data for inflated shape comparison

- Nylon (cap)
- Polyester (carcass)

FE cord component properties updated



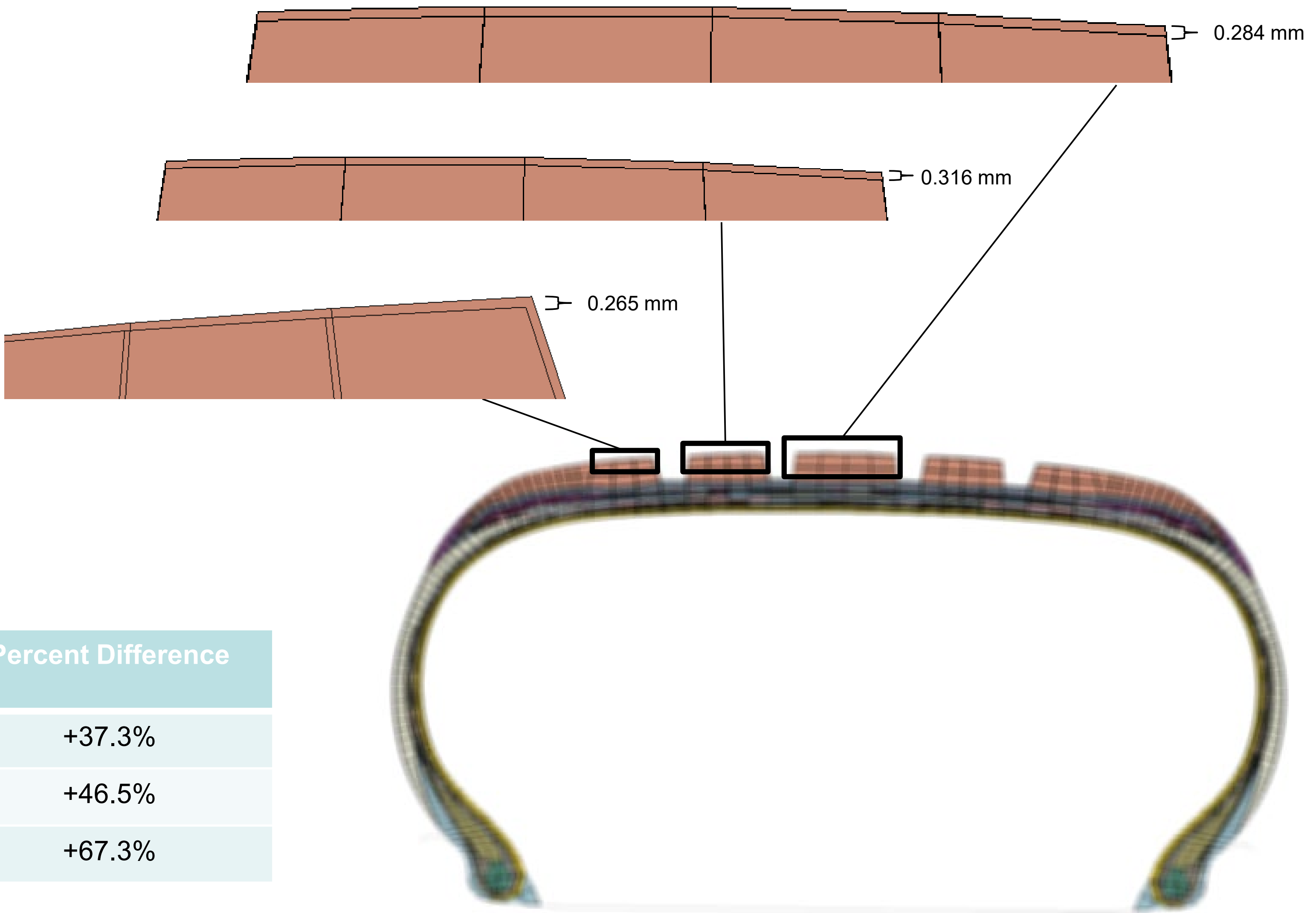
stress-strain properties modified

4) Inflated Tire P-V Analysis

Average improvement of 50.4% at grooves

Hard to tell at sides due to T360 processor limitations and aforementioned intersection

Validate importance of lower strain range in Abaqus/VTA input files



Next steps: Check relationship at other inflations, green angles, mold lift, determine if relationship holds at sides of tire profile using better T360 processor, refine cord database if conclusion further proven

	VTA Deviation	Cord Modified Deviation	Percent Difference
Leftmost Groove	0.71 mm	0.445 mm	+37.3%
Center Left Groove	0.679 mm	0.363 mm	+46.5%
Center Groove	0.422 mm	0.138 mm	+67.3%

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